Lesson 3: Take Your Shot

001 Game Creator (Windows only)

VIDEO GAME CREATION

This week: spawning objects at runtime. Your job is to make your player shoot a projectile that flies up the screen and cleans itself up when it leaves - the first piece of your capstone's actual shooting mechanic.

Open your own [YourName]Game project from Lesson 2 - don't start a new one. Everything this week builds on the player object you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder projectile image (bullet_yellow.png) and a new sound effect (sfx_shoot.wav) - you don't need to find or make your own art today.

Setup

1. Open your [YourName]Game project from Lesson 2.
2. Create a new Actor Template for the projectile (Actor > Dynamic Object template works well) using bullet_yellow.png (import it from this lesson's Assets folder) as its Body sprite, and name it bullet.
3. Place one bullet Actor on the Map, directly on top of main - this placed copy is only there so "Map – Spawn Actor" has a template to copy from; tick "Initially Existent" off for it, so it's invisible and inactive until spawned.
4. On the bullet Actor's Automatic Movement settings, tick "Move" and set its direction to Up, at a comfortable speed - this makes every spawned copy fly upward on its own, no scripting needed for the movement itself.

Level 1 - Minimal (MVP)

Pressing the fire key spawns one projectile that flies upward and disappears when it leaves the screen. This is a complete finish for today - reaching this is enough to move on to next week.

5. Go to Resources > Input Triggers, and add a new Control Combination for the Spacebar, set to "Pressed."
6. Click "Edit Script" for that trigger and add a "Map – Spawn Actor" event, targeting the bullet Actor.
7. On the bullet Actor itself, add an Actor Trigger: "Left / Entered Camera" (the Left case) -> a script with an "Actor – Destroy/Revive Actor" event (Destroy) targeting itself (relate/this).
8. Save, then click Play/Test and press spacebar - a bullet should spawn at its placed position and fly upward, then disappear once it leaves the camera view.

Alternate Path: Textual Scripting (typed BASIC-like code)

001 Game Creator has a second scripting mode that uses typed, BASIC-like commands instead of the graphical event blocks you used above.

IMPORTANT - READ BEFORE YOU TRY THIS: switching a project's Mode from Graphical to Textual is PERMANENT once you save. Do NOT convert your main [YourName]Game project. Create a separate throwaway test project for this exercise only.

9. Create a new, separate project (e.g. [YourName]GameTextual) - do not use your main Semester 1 project for this.
10. Open the Mode menu and switch this new project to Textual (confirm with your instructor that you understand this cannot be undone once saved).
11. Open the Script Editor for a Spacebar Input Trigger, same as before.
12. Instead of dragging graphical event blocks, type the command directly - start typing "Spawn" and use the editor's autocomplete/command list to find the text equivalent of the Map – Spawn Actor event you used in the graphical path.
13. Save and Test to confirm the same result as your graphical version.

Exact command names can vary slightly by version - the Script Editor's autocomplete and the 001gamecreator.com/wiki Textual Scripting reference show the current syntax. If you want to keep exploring this mode, do it in your throwaway test project, not your real one.

Level 2 - Complete (Alpha/Beta)

Firing repeatedly produces multiple independent projectiles on screen at once, each moving and disappearing correctly, and rapid-fire is capped so holding the key down doesn't flood the screen.

- Test firing several times quickly - each press should spawn its own independent bullet, all moving and being destroyed correctly. "Map – Spawn Actor" creates a fresh copy each time it runs, so this already works with no extra scripting.
- Add a cooldown so holding spacebar doesn't fire every frame: Input Triggers set to "Pressed" already only fire once per physical press (not every frame held), so Level 1 is already immune to this specific bug.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: import sfx_shoot.wav (Resources > Import > Sound), then add a "Play Sound" event directly after the "Map – Spawn Actor" event on the fire trigger's script.
- Visual: give the bullet Actor its own Tint/Opacity setting (Actor > Graphics Options) distinct from main's, so it reads clearly as "a shot" rather than a copy of the player.
- Structure: keep everything about how a bullet behaves - its automatic movement setting and its own "Left/Entered Camera" cleanup trigger - defined on the bullet Actor itself, and keep main's Input Trigger script limited to just spawning it (Spawn Actor + sound). Don't add bullet-specific logic to main's scripts, or vice versa.

If Something's Wrong

- Nothing happens on spacebar: confirm the Input Trigger is actually set to "Pressed," and that its script is connected to the Start node in the Script Editor.
- Bullet spawns but doesn't move: confirm "Move" is ticked in the bullet Actor's Automatic Movement window, with a direction and speed actually set (not left at 0).

- Bullet never disappears: confirm the "Left/Entered Camera" trigger is on the bullet Actor itself (not main), and that it's specifically checking for the "Left" case, not "Entered."
- Dialog fields may look slightly different depending on your version - the Script Editor and Actor pages at 001gamecreator.com/wiki have the current layout.

Want More?

001gamecreator.com/wiki, Student Resource List > 001 Game Creator